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TECTONIC ROTATIONS WITHIN THE RIO GRANDE RIFT: EVIDENCE FROM PALEOMAGNETIC STUDIES

Laurie L. Brown

Department of Geology and Geography, University of Massachusetts, Amherst

Matthew P. Golombek

Lunar and Planetary Institute, Houston, Texas
Jet Propulsion Laboratory, California Institute of Technology, Pasadena

Abstract. Paleomagnetic studies on Miocene-Pliocene volcanic rocks from the Espanola basin of the Rio Grande rift, New Mexico, reveal directions discordant from the expected mean direction for North America. The Paliza Canyon Formation, Tschicoma Formation, and Lobato Basalt, all sampled in the Jemez Mountains west of the Pajarito fault zone, have mean declinations east of the expected mean. The Cerros del Rio volcanics, lying east of the Pajarito fault zone, have a westerly declination. Combined with published data on the Santa Fe Group sediments east of the fault zone, and the Valles Rhyolite, west of the fault zone, distinct rotations of the two areas are evident. The western block has rotated clockwise 12°, while the eastern block shows 16° of counter-clockwise motion. Differential rotations of 25° -30° are calculated between the two blocks; 4°/m.y. is the minimum differential rotation for the past 5 m.y. Geologic explanations for these rotations include the opening of the Rio Grande rift in response to clockwise rotation of the Colorado Plateau and significant left slip along the Rio Grande rift.

Introduction

The Rio Grande rift is an intraplate rift that extends from south-central New Mexico to central Colorado. It lies along the boundaries of and appears related in both location and/or evolution to several physiographic provinces: the Great Plains to the east, the Southern Rocky Mountains to the north, the Colorado Plateau to the west, and the Basin and Range to the south. Extensive geological and geophysical work in this area shows that the rift is the site of recent faulting, high heat flow, high seismicity, vertical crustal movements, recent vulcanism, shallow magma bodies, a broad axial uplift, and a thin crust and lithosphere [e.g., Cordell, 1978; Hawley, 1978; Riecker, 1979].

Rio Grande rift extension began about 32 Ma in the south and 27 Ma in the north [Chapin, 1979] in response to a regional stress field that similarly affected much of the Basin and Range province [Zoback et al., 1981]. The rift seems to have broken along an older north trending flaw left over from Laramide (Southern Rocky Mountains) compressional deformation [Chapin, 1979], but extension has not affected the adjacent Colorado Plateau. This has led to the suggestion that the Rio Grande rift opened in response to clockwise rotation of the Colorado Plateau [Eardley, 1962, p. 510; Hamilton and Myers, 1966]. Although this idea has persisted to the present, specific data necessary to substantiate it are only now being presented [Cordell, 1982; Golombek et al., 1983]. Alternatively, or in addition to, several workers have presented evidence for significant left slip resulting in counter-clockwise rotation of crustal blocks within the rift [Kelley, 1979; 1982; Muehlberger, 1979].

In this paper we present new paleomagnetic data from less than 10 m.y. old volcanics within the Espanola basin of the central Rio Grande rift to test for tectonic rotations of crustal blocks. These data support (1) clockwise rotation of rocks in the western part of the Espanola basin, in agreement with models suggesting rotation of the Colorado Plateau, and (2) counterclockwise rotation of rocks within the central Espanola basin, in support of left slip along the Rio Grande rift [Golombek and Brown, 1981; Brown and Golombek, 1982; 1983].

Geologic Setting and Volcanic Stratigraphy

In north-central New Mexico the Rio Grande rift consists of a series of north trending, fault-bounded basins that are asymmetric and step en echelon to the right. These basins are filled with predominantly Miocene sediments of the Santa Fe Group that have accumulated since the onset of rifting. Volcanic activity in this part of the rift is predominantly restricted to several distinct fields, which are from south to north (Figure 1) Albuquerque and San Felipe volcanic fields within the Albuquerque-Belin basin, Cerros del Rio and Jemez volcanic fields within the Espanola basin, and the Taos Plateau volcanic field within the San Luis basin. These rocks overlie the Santa Fe Group and are primarily low-lying Plio-Pleistocene basalt fields except for the Jemez Mountains, which contain basaltic through rhyolitic rocks of late Miocene to Recent age.

The Espanola basin is asymmetric with sediments dipping to the west in accord with major border faults along the western side of the basin. Motion along these faults and other faults throughout the basin probably tilted the Santa Fe Group sediments to the west roughly 7.5 Ma [Golombek et al., 1983]. Present activity is along the central Velarde graben, which began forming in early Miocene time [Manley, 1979]. The western border of the Velarde graben is marked by the pro-

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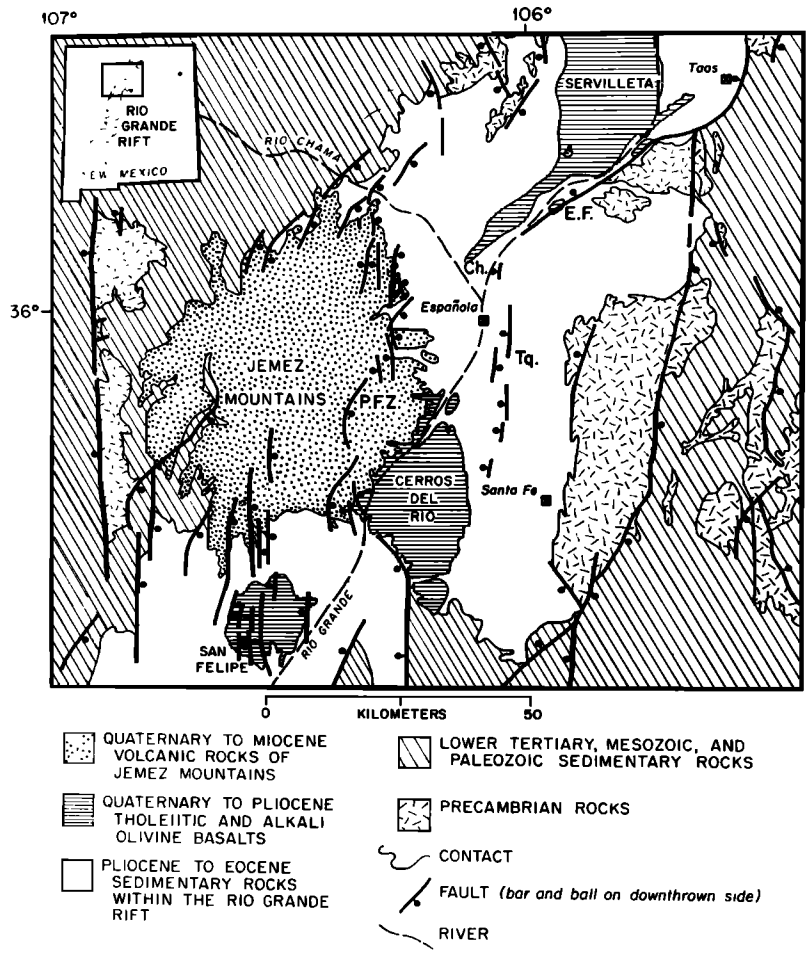


Fig. 1. Generalized geologic map of the Espanola basin, Rio Grande rift, New Mexico. E. F., Embudo fault; P. F. Z., Pajarito fault zone; Ch., sampling locality, Chamita formation: Tq., sampling locality, Tesque Formation.

minent Pajarito fault zone. The Pajarito fault zone cuts the eastern flank of the Jemez Mountains offsetting a 1.1 Ma ash-flow tuff up to 100 m [e.g., Golombek, 1981]. Kelley [1979] defined the Jemez bench west of the Pajarito fault zone as a structural high that did not subside very much during rifting; it was subsequently capped by the Jemez volcanic field.

The oldest volcanism in the Espanola basin is recorded by late Miocene basalt flows interbedded in the sediments of the Santa Fe Group northeast of the Jemez Mountains near Rio Chama [Galusha and Blick, 1971; Manley and Mehnert, 1981]. These volcanic rocks, herein informally referred to as Santa Fe basalts, are dated at 9.9, 9.6, and 8.3 Ma old (see Figure 2 for stratigraphy and sources of dates). They have conformable dips with the overlying and underlying Santa Fe Group and lie in a complexly deformed and tilted region along a possible northeast extension of the Pajarito fault zone.

The earliest Jemez Mountain volcanics, which overlie the Santa Fe Group, are the Keres Group made up predominantly of the Paliza Canyon Formation (8.5-9.1 Ma). This formation, located in the southern part of the volcanic field, is a complex of basaltic, andesitic, dacitic, and rhyolitic flows, tuffs, breccias, domes, and composite volcanoes [Bailey et al., 1969; Smith et al., 1970]. With the eruption of Polvadera Group rocks (Lobato Basalt and Tschicoma Formation), volcanic activity shifted to the north. The Lobato Basalt (7.4-9.6 Ma) forms broad, flat mesas overlying Santa Fe Group sediments. The voluminous Tschicoma Formation (3.7-6.7 Ma) overlies the Lobato Basalt and forms large, highstanding volcanic constructs. The Tschicoma Formation is a complex association of thick latite to quartz latite flows and domes [Bailey et al., 1969; Smith et al., 1970].

The youngest volcanics of the Jemez Mountains are the Tewa Group, which is composed primarily of the Bandelier Tuff and Valles Rhyolite. The Bandelier Tuff consists of two pumaceous, rhyolitic ash flow tuffs which resulted from massive, central caldera-producing explosions 1.4 (Otowi Member) and 1.1 (Tshirege Member) Ma. These ash flows flank the Jemez Mountains in broad, gently dipping plateaus such as the Pajarito Plateau to the east. The last volcanic activity occurred from 0.9 to 0.4 Ma as rhyolitic domes and flows within the Valles Caldera known as the Valles Rhyolite [Bailey et al., 1969; Smith et al., 1970].

The Cerros del Rio volcanic field in the southern Velarde graben is composed of predominantly low-lying basalt flows dotted with cinder

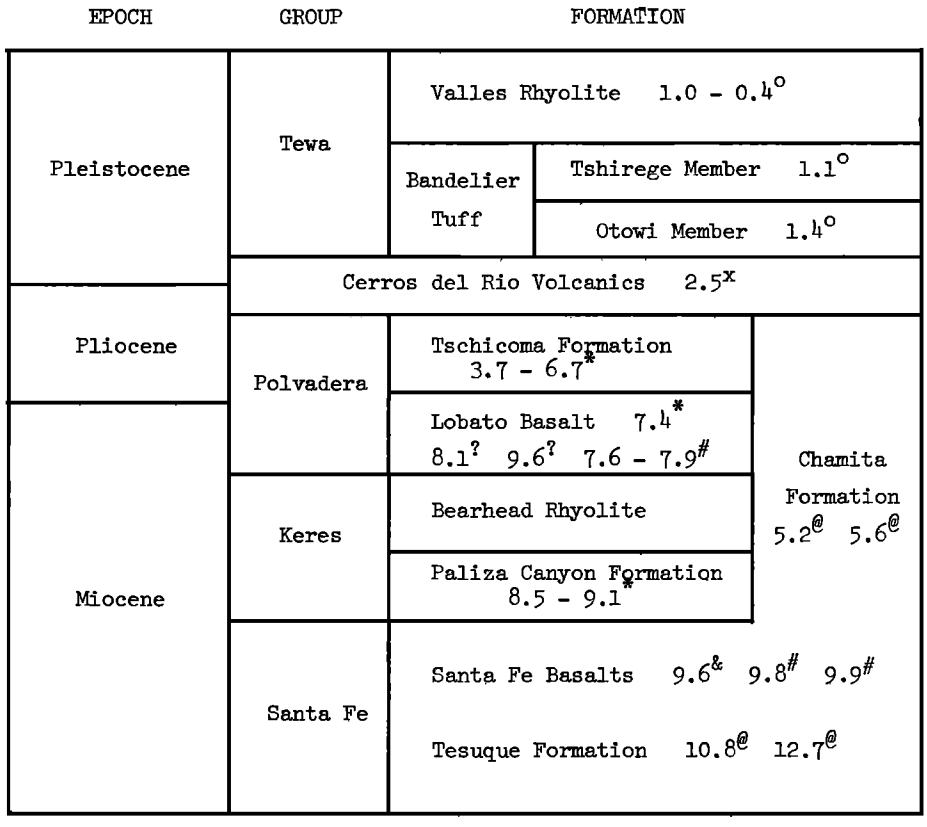


Fig. 2. Stratigraphic column and radiometric ages for rocks in the central Espanola basin. Numbers are ages in millions of years. Data sources: cross, Bachman and Mehnert [1978]; ampersand, Baldridge et al. [1980]; asterisk, Dalrymple et al., [1967]; dot, Doell et al. [1968]; number symbol, Manley and Mehnert [1981]; each symbol, Manley and Naeser [1978]; question mark, unpublished data listed by Luedke and Smith [1978].

cones [Aubele, 1978; 1979]. Radiometric data indicate ages of roughly 2.5 Ma old; however, field relations suggest some volcanism after the deposition of the Otowi Member of the Bandelier Tuff approximately 1.4 Ma [Smith et al., 1970].

Previous Paleomagnetic Studies

Several paleomagnetic studies have been published on other rocks in the Espanola basin. Results of these studies will be used in the following sections on tectonic interpretations and so will be briefly reviewed here.

The upper unit of the Santa Fe Group, the Chamita Formation, was studied by MacFadden [1977] at its stratotype north of Espanola, near the confluence of the Rio Chama and Rio Grande (Figure 1). These clayey silts to medium sands constitute the youngest terrestrial sediments deposited in the rift valley. Biochronologic studies on this sequence indicate deposition during late Hemphillian time (late Miocene-early Pliocene) [MacFadden, 1977]. Fission track dates on two interlayered ash beds yield ages of 5.6±1.0 Ma and 5.2±1.0 Ma [Manley and Naeser, 1977]. Twelve magnetic polarity zones are recognized in the Chamita Formation; they correlate to the magnetic time scale from Epoch 5 through the Gilbert Epoch [MacFadden, 1977]. The mean magnetic direction for the stratotype is inclination = 48.8° and declination = 345.8° with N = 106 and alpha-95 = 3.1°, which MacFadden noted is divergent from the axial dipole field direction.

The magnetostratigraphy of the lower part of the Santa Fe Group, the Tesuque Formation, has been extensively studied by Barghoorn [1981]. The middle to late Miocene Nambe, Skull Ridge, and Pojoaque members were sampled in their type localities in the Espanola basin (Figure 1). The resulting magnetic polarity sequence correlates with Epoch 16 through Epoch 12. Reversed and normal polarity sites are closely anti-parallel, but polarity means are slightly statistically different from each other. This is attributed by Barghoorn to incomplete magnetic cleaning of some of the reversed sites. For this reason, correlations discussed here use the mean direction from normal sites only (inclination = 48.7° and declination = 338.8°, N = 49, alpha-95 = 3.7°) (S. Barghoorn, personal communication, 1982).

A classic paleomagnetic study by Doell et al. [1968] investigates the Valles Rhyolite and the Bandelier Tuff. A magnetic stratigraphy of reversed-short normal-reversed-normal corresponds to Matuyama through Brunhes Epoch and the first identification of the Jaramillo event. The oldest dated volcanic units studied by Doell and coworkers are the upper and lower members of the Bandelier Tuff, with ages of 1.1 and 1.4 Ma, respectively. Using data given by Doell et al. [1968], a mean direction of inclination = 51.2° and declination = 5.3° (N = 15 and alpha-95 =

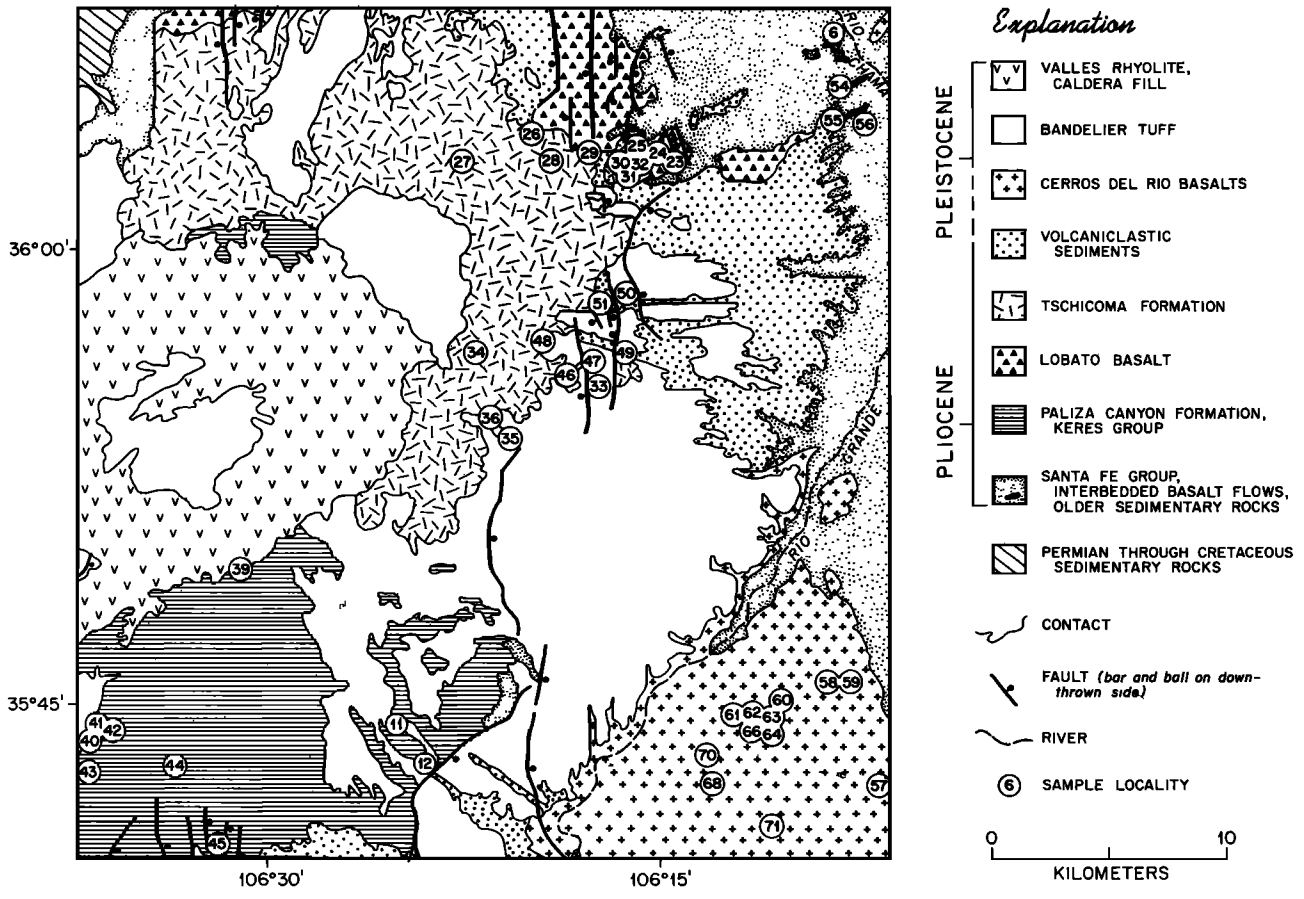


Fig. 3. Geologic map of the eastern part of the Jemez Mountains, western Espanola basin (modified from Smith et al., [1970]). Circled numbers refer to paleomagnetic sites of units discussed in text.

5.8°) is obtained for the Valles Rhyolite and the Bandelier Tuff, excluding one site with an intermediate direction.

Paleomagnetic Techniques

Samples from the Lobato Basalt, Tschicoma Formation, Paliza Canyon Formation, Santa Fe basalts, and Cerros del Rio volcanics were collected during the summers of 1979 and 1980 as part of an investigation of the Espanola basin rift structure and tectonics [Golombek et al., 1983]. All units were drilled in the field using a gasoline-powered rock drill [Doell and Cox, 1967]. Both magnetic and sun compasses were used to orient samples; differences between the two were rarely greater than 2°. Site locations (Figure 3) were chosen to provide broad coverage of the units involved and to investigate their relationship to the Pajarito fault zone. Local geology was determined from mapping by Golombek [1983] and other published geologic maps of the area by Smith et al. [1970], Kelley [1978], and Aubele [1978].

Each site consists of five to eight drilled cores; single samples 2.4 cm in diameter and 2.4 cm in length were cut from each core. All samples were measured on a Schonstedt Spinner Magnetometer, with alternating field demagnetization studies done using a single axis Schonstedt demagnetizer. In most cases, two samples from each site were subjected to alternating field demagnetization in steps of 2.5, 5.0, 10, 20, 30, 40, 60, and 80 mT. Mean coercivities were variable, and all additional samples were demagnetized at 10, 20, and 40 mT. Site means listed in Table 2 were calculated at demagnetization levels with the least scatter.

Geologic evidence indicates that tilting in the Espanola basin occurred approximately 7.5 Ma ago [Golombek, 1983; Golombek et al., 1983]. This event affected Santa Fe sediments, the Paliza Canyon Formation and basal Lobato Basalt. Although tilting was predominantly to the west in the central and eastern parts of the basin, the Jemez Mountains do not reveal consistent regional tilts. Rather small eastward as well as small westward tilts (0°-10°) are observed in scattered locations in the Paliza Canyon Formation [Smith et al., 1970] and the Lobato Basalt [Smith et al., 1970; Kelley, 1978]. It is noted that if a consistent westward tilt is present, removal would shift declinations westward. Likewise, if a consistent eastward tilt exists, as is suggested by Smith et al., [1970, x-section AA'], correction would produce more easterly declinations.

Owing to the massive nature of the units studied here, and the lack of observable contacts or other indicators of original horizontal, it was impossible to make site by site tilt corrections.

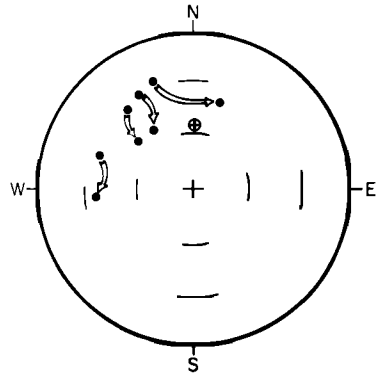


Fig. 4. Equal-area plot of demagnetized site mean directions, Santa Fe basalts, before and after tectonic correction. Arrows indicate direction change after correction by simple rotation around the local strike. Solid circles indicate lower hemisphere directions; circled cross is dipole field for site latitude.

Also, study mean directions could not be corrected for a regional tilt, as it is not consistent. Although some of the scatter in the Paliza Canyon and Lobato data may be due to uncorrected tilts, the presence of small eastward and small westward dips in the study area is confusing. If these opposing tilts could be removed, it is probable that the precision would increase, but the mean direction remain unchanged.

The younger Tschicoma Formation is essentially flat-lying, showing only localized tilting around recently active faults. Some of the Cerros del Rio volcanics may have flowed down a gentle gradient to the north [Aubele, 1979], but again site by site corrections could not be made.

Results

Santa Fe Basalts

Four basalt flows (≈10 Ma) interbedded in the sediments of the upper Santa Fe Group northeast of the Jemez Mountains were sampled in the area of Chili, New Mexico (Figure 3, sites 6, 54, 55, and 56). Mean directions for each site after demagnetization at 20 mT, are shown in Figure 4, with statistics listed in Table 1. The strike and dip of underlying sediments were used to untilt the basalts by simple rotation around the local strike. As is evident from Figure 4, these sites cluster better before tectonic corrections than after, with k decreasing from 18 to 7. This implies a more complex history of tilting and rotation which cannot be removed at this time.

Paliza Canyon Formation

The Paliza Canyon Formation, 8.9-9.1 Ma, was sampled in the southern part of the Jemez Mountains. Sites were located in basalts (39, 41, and 42), andesites (12, 40, 43, and 45), and dacites (44) as mapped by Smith et al., [1970]. One site (11) from the Bearhead Rhyolite of similar age is included with the Paliza Canyon results. As indicated by the natural remanent magnetization (NRM) data there are both normal and reversed components present. Demagnetization at 20 mT does not alter the mean directions but does decrease within-site scatter. The basalts and andesites have both polarity directions; the dacites are all normal.

Although the within-site scatter is low (Table 2), there is considerable between-site scatter (Figure 7c). Sites 12 and 45 were collected immediately adjacent to major faults, and both probably show considerable related tilting. Since adequate field information was not available to allow for structural corrections, these two sites have been omitted from further calculations. Mean directions for the formation, with the reversed sites inverted, are inclination = 51.1° and declination = 9.2°, with N = 7, alpha-95 = 11.3°.

Lobato Basalt

The Lobato Basalt (approximately 7.5 Ma) was sampled at nine sites, seven sites from the northeastern part of the Jemez Mountains and two sites north of Figure 3. NRM directions are nearly all normal (Figure 5), and most samples behaved in a stable manner during step demagnetization (Figure 6). There is little indication of secondary components, and mean coercivities for most samples were about 20 mT. The two northern sites were very scattered after demagnetization (alpha-95 > 30°) and have been omitted.

TABLE 1. Magnetic Directions, Santa Fe Basalts

Site	N	Uncorrected I	Uncorrected D	Corrected I	Corrected D	K	a-95	Pole °N	Pole °E
6	7	31.4	328.8	51.4	328.4	30	11.2	63.6	164.3
54	6	27.9	339.8	41.3	17.1	61	8.6	70.8	19.1
55	7	34.0	321.8	51.3	312.2	260	3.7	50.6	172.2
56	6	36.3	287.6	37.5	267.6	36	11.3	10.3	182.3
MEAN BEFORE TECTONIC CORRECTION	4	33.9	320.4			18	22.0	59.9	171.0
MEAN AFTER TECTONIC CORRECTION	4			53.1	320.2	7	38.9		

Notation as in Table 2.

TABLE 2. Magnetic Directions, Mio-Pliocene Units

Site	N	I	D	K	a-95	Pole °N	Pole °E
		Cerros del Rio Volcanics					
57	4	-61.4	152.2	131	8.0	67.5	189.9
58	6	-48.6	84.2	138	5.9	12.5	191.4
59	5	-61.7	176.1	248	4.9	82.2	231.9
60	4	36.9	294.9	11	29.3	31.7	167.6
61	6	51.5	351.9	10	22.6	82.4	138.7
62	4	-57.3	171.1	15	24.5	82.5	183.5
63	5	-50.0	183.2	17	18.9	84.5	44.2
64	5	-64.9	138.4	48	11.2	57.3	196.8
66	6	-66.3	156.8	355	3.6	69.2	205.1
68	6	12.6	14.2	9	23.9	57.9	46.5
70	6	51.5	353.1	174	5.1	83.3	134.3
71	6	27.3	351.0	647	2.6	67.3	96.9
Mean	12	53.6	340.2	10	14.4		
				8	16.4	73.3	170.4
		Tschicoma Formation					
26	6	-27.7	188.8	32	12.1	67.3	51.2
27	5	37.7	352.9	25	15.8	73.9	98.1
28	6	-46.3	177.7	110	6.4	81.4	87.4
29*	6	8.3	274.4	77	7.8	6.0	164.5
33	7	64.4	21.3	142	5.1	71.0	304.3
34	6	38.6	40.7	45	10.1	52.0	354.2
35	6	52.1	356.9	101	6.7	85.9	113.7
36	6	-53.4	172.9	333	3.7	83.9	147.3
46	7	71.3	37.6	140	5.1	57.7	293.5
47	6	38.8	46.7	32	12.1	47.2	349.7
48	6	51.4	12.2	29	12.7	79.2	1.1
49	4	46.6	32.2	512	4.1	61.6	351.0
51	6	-52.6	207.8	22	14.5	67.0	342.5
Mean	12	50.0	16.7	21	9.7		
				16	11.2	75.6	354.2
		Lobato Basalt					
23	6	59.0	29.0	705	2.5	66.9	315.6
24	5	36.9	325.2	16	19.6	73.1	99.6
25	4	36.8	358.7	13	26.1	74.4	78.3
30	4	48.6	5.3	36	15.6	82.1	37.7
31	6	63.1	.11.0	164	5.2	78.0	294.7
32	6	60.0	25.7	441	3.2	69.4	322.3
50	6	45.4	15.1	236	4.4	74.3	14.7
Mean	7	51.8	4.5	21	13.5		
				29	11.4	81.0	4.1
		Paliza Canyon Formation					
11	7	-46.0	191.6	22	13.1	72.0	20.8
12*	6	14.8	319.4	19	14.0	43.3	136.4
39	6	-58.0	225.1	57	8.9	54.3	325.1
40	6	-34.0	178.9	254	4.2	72.9	76.9
41	5	-49.6	208.0	52	10.7	66.0	348.1
42	7	50.4	357.8	199	4.3	85.1	95.9
43	7	-53.5	161.8	88	7.2	75.0	162.3
44	5	55.9	9.2	108	7.4	82.5	335.0
45*	4	-18.2	229.8	106	8.9	37.7	1.1
Mean	7	51.1	9.2	29	11.3		
				19	14.2	80.5	4.5

N, number of samples per site; I and D, magnetic inclination and declination in degrees; K, precision parameter; a-95, radius of circle of 95% confidence about the mean direction; pole is north VGP in degrees north latitude and east longitude. Mean directions calculated after inverting reversed sites.

*Site excluded from study mean.

Average data for all other sites are given in Table 2. As can be seen from Figure 7b, there is considerable agreement between sites except for site 24. This may be due in part to the choice of that sampling site immediately adjacent to a mapped fault. Within-site scatter is generally small, although larger scatter is seen in sites 24, 25, and 30. Sampling sites are on a series of fault blocks with both down to the east and down to the west motion, and no correction for tilt was attempted. Unit mean directions for the seven sites listed in Table 2 are inclination = 51.8° and declination = 4.5°, for N = 8 and alpha-95 = 13.5°.

Tschicoma Formation

Fifteen sites in 3.7 to 6.7 Ma quartz latites of the Tschicoma Formation were sampled in the central and northern parts of the Jemez Mountains (Figure 3). NRM directions indicating normal, reversed, and intermediate directions are present (Figure 5). Detailed demagnetization studies on selected samples from each site showed little or no variation in direction with increasing alternating fields. A typical Zijderveld diagram is shown in Figure 6. Average mean coercivities for samples treated in this manner were between 20 and 40 mT, and all remaining samples were demagnetized at both 20 mT and 40 mT levels. Two sites, one from the northernmost part of the area (off the map of Figure 3) and one from the southern part of the area gave very scattered results and have not been considered in the statistical analysis. Of the 13 remaining sites, eight have normal directions, four are reversed, and one is intermediate with a shallow inclination (Figure 7a). Normal and inverted reverse sites are similar at the 95% level. Without the two scattered sites and the intermediate site, the mean direction for the Tschicoma Formation is inclination = 50.0°, declination = 16.7°, N = 12, and alpha-95 = 9.7° (Table 2).

Cerros del Rio Volcanics

The Cerros del Rio volcanic field (≈2.5 Ma) lies southeast of the Jemez Mountains at the southern end of the Espanola basin. Twenty sites were sampled throughout this field in flows of hawaiite, andesite, and basaltic andesite. Alternating field step demagnetization to 100 mT on selected samples from each site revealed little direction change and little or no viscous overprinting. Despite this stable decay behavior, eight sites yielded very scattered directions (alpha-95 > 30°) at 40 mT. These sites all have NRM intensities an order of magnitude greater than the remaining 12 sites, and lightning strikes are suspected. Although intensity decays with increasing demagnetization in these scattered sites, with 10-20% left at 40 mT, there is no accompanying direction change. These eight sites have been omitted from Table 2 and are not considered further. The remaining 12 sites appear stable, and over half of them show little within-site scatter (Table 2). There are five sites listed with smaller precision parameters (k = 9-17) than usual for such young rocks. The reason for such poor data is not evident and may be due to a number of factors. All sites are from natural out-

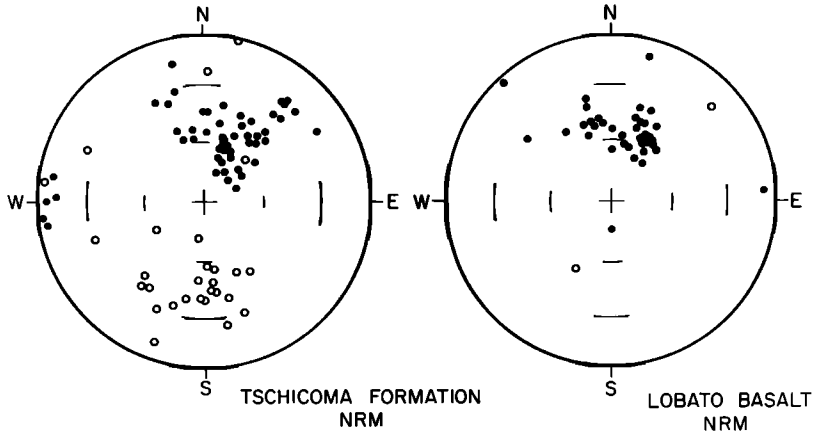


Fig. 5. Equal-area plot of natural remanent magnetization of all samples from the Tschicoma Formation and the Lobato Basalt. Solid circles indicate lower hemisphere directions; open circles are upper hemisphere directions.

crops, often flow edges, and may be effected by disturbances during cooling. Nevertheless, several of these flows (61, 62, 63) have directions very similar to the high-precision sites. Because of this and the already small number of usable flows, it was decided to use the low-precision sites in determining a mean direction.

Mean directions for the acceptable sites are plotted in Figure 8, with five of the sites having normal polarity and seven sites being reversed. Between-site scatter for the normal polarity sites is large, and all normal directions have inclinations shallower than the axial geocentric dipole field for this latitude. The reversed sites are well clustered and generally show steeper inclinations, except for one site with a transitional direction. Mean direction for the Cerros del Rio volcanics after inverting the reversed sites, is inclination = 53.6° and declination = 340.2° for N = 12 and alpha-95 = 14.4°.

Rotations

Mean magnetic directions for units studied here and those available in the literature plot close to, but not coincident with, the mean direction for North America in the late Tertiary (Figure 9). In this case a mean pole for North America was calculated using late Tertiary data listed by Irving and Irving [1982, Table 1]. Paleopoles 1-12 with ages less than 10 Ma were used, omitting pole 5 (Servilleta basalt flows from the Rio Grande rift north of the Espanola basin) and pole 9 (Chamita Formation). The mean pole determined from this data set (N=10) is 86.5°N, 114.7°E, with alpha-95 = 3.9°. For the latitude of the Espanola basin, this pole gives a corresponding direction of inclination = 52.7° and declination = 357.3° (alpha-95 = 3.5°).

Even more noticeable in Figure 9 is the geographic division of directions into two groups: one with northeasterly declinations, the other

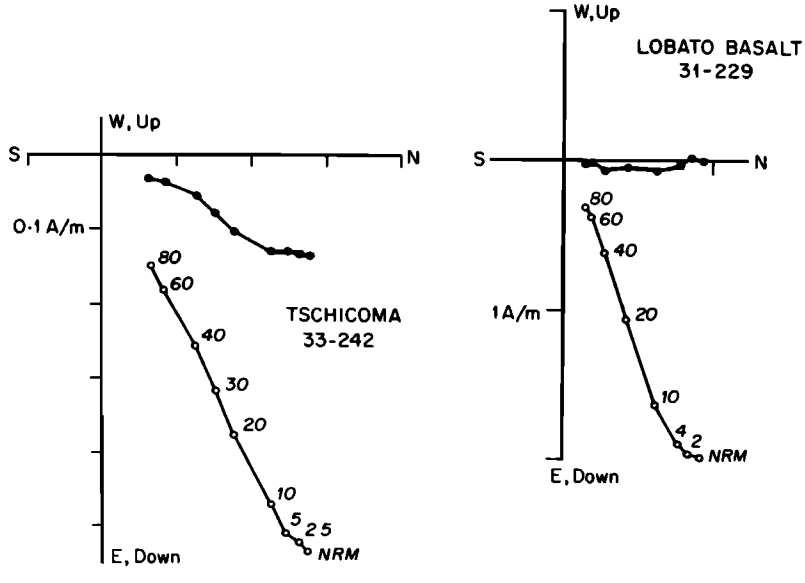


Fig. 6. Typical demagnetization diagrams for a representative sample each from the Tschicoma Formation and the Lobato Basalt. Solid circles are horizontal projections of the demagnetization path; open circles are vertical projections. Alternating field demagnetization levels are in milliteslas.

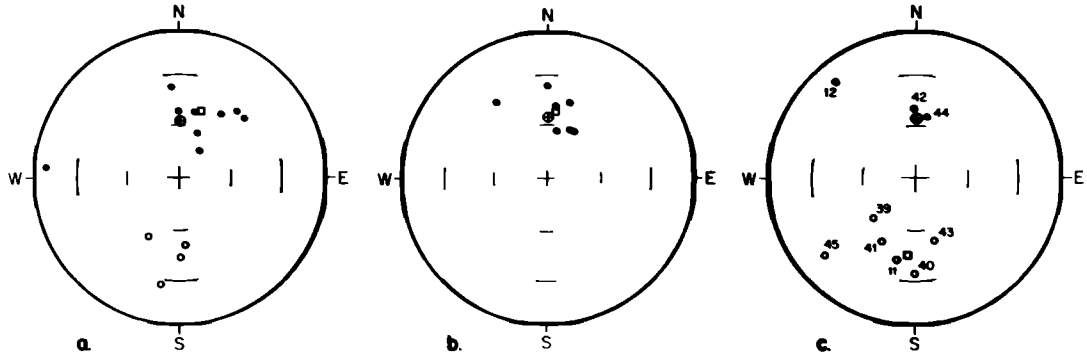


Fig. 7. Equal-area plots of demagnetized site means for units in the western Jemez Mountains. (a) Tschicoma Formation, (b) Lobato Basalt, (c) Paliza Canyon Formation. Numbers refer to site locations in Figure 3. Symbols as in Figure 5, with open square indicating formation mean.

with northwesterly declinations. Those units east of the Pajarito fault zone (Chamita and Tesuque formations of the Santa Fe Group and the Cerros del Rio volcanics) all have northwesterly declinations, while units west of the Pajarito fault zone (Paliza Canyon Formation, Tschicoma Formation, Lobato Basalt, and Valles Rhyolite) all have declinations northeast of the mean field.

Considering these units as two separate groups based on location, their means can be plotted with respective alpha-95 circles of confidence (Figure 10, Santa Fe basalts omitted). The group east of the Pajarito fault zone has mean declination = 341.6°, inclination = 50.4° and alpha-95 = 5.6°. Data for the units west of the fault zone yield a mean direction of declination = 9.0°, inclination = 51.1° and alpha-95 = 4.1°. Using the late Tertiary direction discussed above for comparison, with alpha-95 = 3.5°, the distinct nature of these two populations is clear. Not only are the two groups separate from each other, but they are also distinct from the average direction for North America. This difference is substantiated by a comparison of mean directions test [Watson, 1956], which the data fails, indicating that the directions are significantly different.

Rotation Statistics

To quantify these observations, statistics developed to identify rotations and flattenings in tectonically active areas [Beck, 1980] have been employed. These initial statistics have been used and modified by many workers [e.g., Magill et al., 1981] studying microplate terrain in the Pacific Northwest.

The degree of rotation R clockwise about a horizontal axis and the flattening F about a vertical axis are determined from the relationships

$$R = D_{obs} - D_{exp} \quad (1)$$

$$F = I_{exp} - I_{obs} \quad (2)$$

where declination D and inclination I are compared from the observed unit (obs) and the expected field direction for that age and latitude (exp). Possible errors on these motions are controlled ultimately by alpha-95 values for the units in question, such that

$$\Delta R = ((\Delta D_{obs})^2 + (\Delta D_{exp})^2)^{1/2} \quad (3)$$

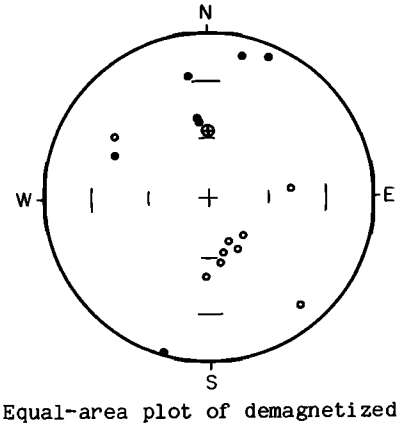


Fig. 8. Equal-area plot of demagnetized site mean directions for Cerros del Rio volcanics. Symbols as in Figure 5.

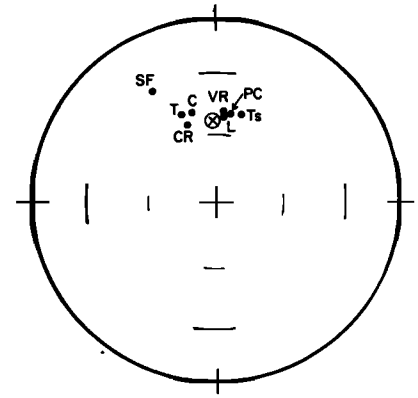


Fig. 9. Plot of mean directions for all formations in the Espanola basin for which paleomagnetic data exists. C, Chamita Formation; CR, Cerros del Rio volcanics; L, Lobato Basalt; PC, Paliza Canyon Formation; SF, Santa Fe basalts; T, Tesuque Formation; Ts, Tschicoma Formation; VR, Valles Rhyolite; circled cross is mean field direction for the late Tertiary. See text for sources of studies. Note that in initial presentation of this data [Golombek et al., 1983, Figure 6], VR and PC were incorrectly plotted.

TABLE 3. Rotation Statistics from Late Tertiary Units, Espanola Basin

Study	Source	N	D	I	a/A		Pole Lat	Pole Long	F	ΔF	R	ΔR	ΔR*
Valles Rhyolite	1	15	5.3	51.2	5.8	6.7	84.6	18.0	1.5	8.4	8.0	9.2	7.3
Tschicoma	2	12	16.7	50.0	9.7	11.2	75.6	354.2	2.7	13.2	19.4	13.9	11.0
Lobato Basalt	2	7	4.5	51.8	13.5	11.4	81.0	4.1	0.9	12.9	7.2	14.3	---
Paliza Canyon Formation	2	8	9.2	51.1	11.9	14.2	80.5	4.5	1.6	16.0	11.9	17.4	---
Cerros del Rio volcanics	2	12	340.2	53.6	14.4	16.4	73.3	170.4	-0.9	17.4	-17.1	20.5	---
Chamita Formation	3	106	345.8	48.8	3.1	---	---	---	3.9	5.1	-11.5	6.6	5.3
Tesuque Formation	4	49	338.8	48.7	3.7	---	---	---	4.0	5.5	-18.5	7.3	5.8
Mean pole			357.3	52.7	3.5	3.9	86.5	114.7					

N, number of sites used for means; D and I, declination and inclination in degrees; a/A, radius of cone of 95% confidence; a, about direction mean; A, about pole mean; pole in degrees north latitude and east longitude; F, ΔF, R, ΔR, flattening, rotation, and corresponding errors calculated from equations given in text; Sources: 1, Doell et al. [1968]; 2, this paper; 3, MacFadden [1977]; 4, Barghoorn [1981]; direction for mean pole calculated for study site.
*Error from Demarest [1983], valid for alpha less than 10° only.

$$\Delta F = ((\Delta I_{exp})^2 + (\Delta I_{obs})^2)^{1/2} \quad (4)$$

$$\Delta D = \sin^{-1}(\sin A\text{-}95/\sin p) \quad (5)$$

or

$$\Delta D = \sin^{-1}(\sin a\text{-}95/\cos I_{obs}) \quad (6)$$

$$\Delta I = 2A\text{-}95/(1 + 3\cos^2 p) \quad (7)$$

or

$$\Delta I = a\text{-}95 \quad (8)$$

where p is the colatitude, A-95 is the cone of 95% confidence about the mean pole, and a-95 is the cone of 95% confidence about the mean direction. A-95 is used whenever possible; if it is not available, a-95 is used.

Demarest [1983] suggests that the above error bar calculation is incorrect and shows analytical calculations resulting in error bars 75% - 80% of those previously determined. This method, however, is only valid for small alpha-95 values (less than 10°) and inclinations distinct from the vertical.

Rotations between separate units, either of similar age indivergent locations or different ages in the same location, can be evaluated by

$$^{A}R_{B} = R_A - R_B \quad (9)$$

$$\Delta_{A}R_{B} = (\Delta R_A^2 + \Delta R_B^2)^{1/2} \quad (10)$$

where $^{A}R_{B}$ is the relative rotation between two units (or blocks) A and B and $\Delta_{A}R_{B}$ is the corresponding error [Magill et al., 1981].

Espanola Basin Data

Rotation statistics for the Miocene-Pliocene rock units from this study and those calculated by us for the other paleomagnetic studies in the area are shown in Table 3. All motions are calculated relative to the mean late Tertiary direction discussed above. The traditional error calculation ΔR is shown for all units, and the Demarest error correction ΔR* is shown for those units with alpha-95 values less than 10°. Although there is a very slight flattening of this data (see Figure 9), none of it is significant at the 95% level. The absence of any significant flattening suggests that uncorrected large tilts are not present, as consistent tilts of 30° would produce observable shallowing of inclinations. The four units west of the Pajarito fault zone all show clockwise (positive) rotations of 7° - 19°. The Valles Rhyolite and Tschicoma Formation are distinct from the reference direction, with rotations of 8° and 19°, respectively. Although the Lobato Basalt and Paliza Canyon Formation have similar rotations (7° and 12°), they have large error bars and are not statistically significant

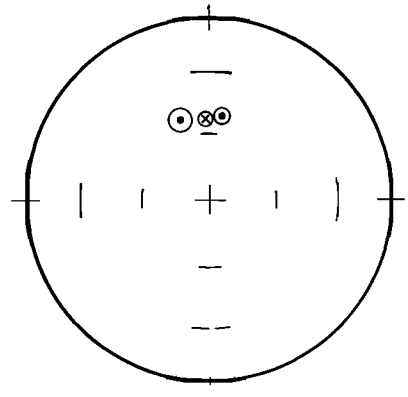


Fig. 10. Equal-area plot of mean directions for units east of the Pajarito fault zone (N=3), units west of the Pajarito fault zone (N=4), and mean direction for late Tertiary, all with circles of 95% confidence. See Figure 9 and text for data used; note that Santa Fe basalt data are omitted.

TABLE 4. Differential Rotation, by Age, Across the Pajarito Fault Zone

Age	Units West	Units East	$_WR_E$	$_W\Delta R_E$	Min $_WR_E$
0-2.5	Valles Rhyolite	Cerros del Rio	25	22	3
5	Tschicoma	Chamita	31	12	19
10	Paliza Canyon	Tesuque	30	18	12

Age in millions of years, $_WR_E$ and $_W\Delta R_E$ calculated from data in Table 3.

at the 95% level. This is in part due to their large alpha-95 values, which directly affect the error calculations. Comparison of the mean direction for these four units to the expected direction yields a rotation of 12° ± 8° for the western block. Thus even though two of the separate studies do not show significant rotations, the western group as a whole shows 12° of clockwise rotation.

The Cerros del Rio volcanics, Chamita Formation, and Tesuque Formation lying east of the Pajarito fault zone all show counterclockwise (negative) rotations of -12° to -19°. For the Santa Fe Group sediments, minimum rotations (R - ΔR) range from -6° to -13° and show larger rotations for the older rocks (Tesuque Formation, R = -18.5°; Chamita Formation, R = -11.5°). The Cerros del Rio volcanics, although they show a relatively large rotation of -17°, prove not to be distinct from the expected direction at the 95% level (i.e., ΔR = 20.5°). Comparison of the mean direction for the three units east of the Pajarito fault zone to the expected direction shows a rotation of -16° ± 10°.

It is possible to look at differential rotations between units of similar age on opposite sides of the fault zone. For this compilation, three average ages were used: 0-2.5 Ma, 5 Ma, and 10 Ma. The rotations, shown in Table 4, range from 25° to 31°, and all are distinct at the 95% level. Minimum possible rotation is 3°, between the Valles Rhyolite and the Cerros del Rio volcanics; minimum rotations of approximately 20° are seen between the Tschicoma and Chamita formations. Again these differential rotations reflect the ages: youngest units have the smallest rotation, older units have larger rotations.

Discussion

Statistically significant clockwise and counterclockwise rotations are evident for rocks in the Espanola basin. Rocks with opposing senses of motion, separated by the Pajarito fault zone, suggest two distinct crustal blocks within the Rio Grande rift that are rotating in opposite directions. Existing geologic and tectonic information, as discussed below, supports such an intrarift model. The eastward difference in declination (with respect to the expected direction) of rocks to the west of the Pajarito fault zone implies that these units have been rotated clockwise approximately 12°. Clockwise rotation of the Colorado Plateau to form the Rio Grande rift has been proposed by Eardly [1962, p. 610], Livacarri [1979], and Hamilton [1981]. Hamilton [1981] calculates 3° of clockwise rotation of the Colorado Plateau. Cordell [1982] and Eaton [1979] similarly suggest clockwise opening of the Rio Grande rift about a pole in Colorado (Figure 11). These suggestions are in agreement with (1) WNW-ESE extensional stress regime throughout the western United States and Rio Grande rift in the past 10 m.y. [Zoback et al., 1981; Golombek et al., 1983], (2) southward widening of the Rio Grande rift requiring greater extension to the south [e.g., Woodward, 1977; Golombek et al., 1983; Brocher, 1981], and (3) the earlier onset of rifting to the south [Chapin, 1979]. Thus we can conclude that there is good evidence in support of a tectonic system involving a few degrees of clockwise rotation of rocks to the west of the Rio Grande rift. It appears that rocks on the Jemez bench, a region west of the Pajarito fault zone in the Espanola basin, rotated in a similar direction.

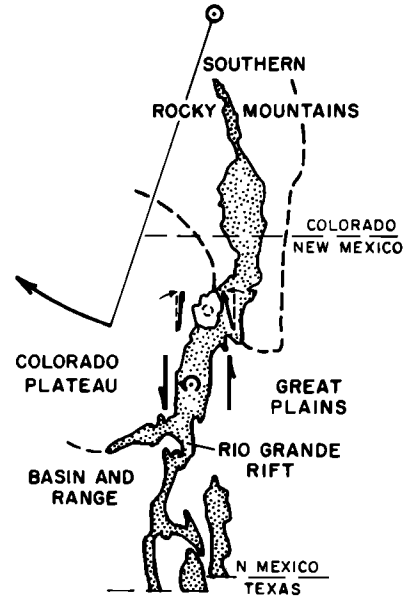


Fig. 11. Schematic diagram of the Rio Grande rift from the New Mexico/Texas border northward into Colorado, with major physiographic provinces labeled. Clockwise rotation of the Colorado Plateau and opening of the rift about apole in northern Colorado (location schematic) indicated, along with a component of left slip along the rift. Small vectors in the region of the Espanola basin indicate the sense of rotation seen in the paleomagnetic declinations.

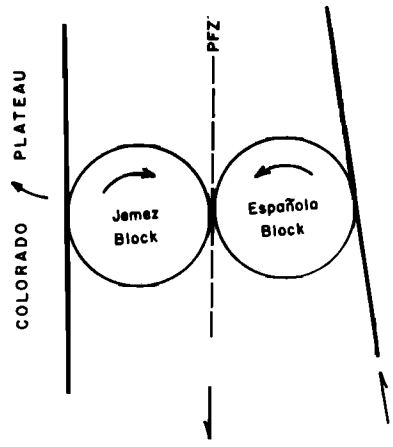


Fig. 12. Schematic cartoon of possible block rotations in the Espanola basin. The Colorado Plateau and Jemez block (western Espanola basin) rotate clockwise. The central and eastern Espanola basin (Espanola block) rotate counterclockwise from left slip along the rift. The blocks are separated by the Pajarito fault zone (PFZ). No scale is implied as the rotating blocks may be small in areal extent.

All rocks studied within the rift east of the Pajarito fault zone have a consistent declination variation of 12°-19° west from the expected direction and, as a group, show 16° of counterclockwise rotation within the rift. The counterclockwise rotation indicated by the data is similar in magnitude to the clockwise rotation of the western part of the rift; it is impossible at this time to determine if this is independent of, or superposed on, the previously discussed clockwise rotation. There is considerable geologic evidence to support left slip (and thus counterclockwise motions) along the Rio Grande rift. Kelley [1977; 1979; 1982] suggests that left slip along the rift is indicated by drag along a number of border faults and associted folds or flexure zones which fit into a general model of en echelon ramping depressions. Furthermore, the southern terminations of Jurassic sedimentary strata on either side of the rift imply roughly 30 km of left slip along the rift [Kelley, 1979; 1982]. Kelley's idea of left slip has been supported by detailed mapping in the northeastern Espanola basin [Steinpress, 1981] and the San Luis basin [Muehlberger, 1979]. Steinpress [1981] documents a component of left slip across a number of faults as indicated by slickensides and displacements. Muehlberger [1979] presents geologic and tectonic evidence for ongoing compression in a NW-SE direction across the Embudo fault. He interprets these data to indicate counterclockwise rotation of a large crustal block bounded by the Embudo fault to the north and the Pajarito fault zone to the northwest. Paleomagnetic data presented here support this interpretation and indicate that the Pajarito fault zone is the boundary between the clockwise rotated western block of the rift and the counterclockwise rotated eastern block of the rift.

The data of differential rotations with age (Table 4) allow us to speculate on the time and rate of rotation. Because the Pajarito fault zone appears to be the boundary between these two blocks, rotation may be no older than the fault zone itself. Golombek [1983], from detailed work in the area, puts the initiation of the Pajarito fault zone at 5 Ma. Rotations beginning at this time seem likely, especially since the older (10 Ma) Paliza Canyon Formation-Tesuque Formation rotations are not larger than the 5 Ma Tschicoma Formation-Chamita Formation rotations. Using the minimum rotations implied by this latter pair (20°), we can calculate an average minimum differential rotation of approximately 4°/m.y. This agrees well with the minimum rotation for the youngest pair, Valles Rhyolite-Cerros del Rio, of 3°. Minimum values are used to present the most conservative possibility; the average differential rotation is considerably larger.

Although the rotations indicated by the paleomagnetic data are large, a number of considerations suggest that such rotations may not be unreasonable, given the possible subsurface conditions beneath the Espanola basin. Rocks located to the west of the Pajarito fault zone on the Jemez bench (informally referred to as the Jemez block in Figure 12) are rotated in a clockwise sense in accord with clockwise rotation of the Colorado Plateau and/or opening of the Rio Grande rift about a pole in northern Colorado. Estimates of the opening angle for the rift are smaller (e.g., Cordell [1982] calculates 1.3°) than the 12° rotation indicated by the paleomagnetic data. It is important to recognize, however, that a large magma chamber probably underlies much of the Jemez Mountains at depths of only a few kilometers [e.g., Kolstad and McGetchin, 1978]. Given both the WNW-ESE extension direction in the last 10 m.y. [Zoback et al., 1981; Golombek et al., 1983] and the probable existence of liquid magma at shallow depths for much of the last 10 m.y., it is possible that small crustal blocks (essentially detached or "floating" at depth) within the Jemez Mountains have rotated clockwise (in response to the WNW-ESE extension direction) by a greater amount than the adjacent Colorado Plateau. Rocks located to the east of the Pajarito fault zone (referred to informally as the Espanola block in Figure 12) are rotated in a counterclockwise sense, probably due to left slip along the Rio Grande rift. In this case, the amount of rotation is dependent on the amount of left slip and the size of the rotating block. Judging by the size of the large rhomb-shaped block proposed by Muehlberger [1979] (bounded to the north by the Embudo fault and to the northwest by the Pajarito fault zone), only a couple of degrees of rotation could occur without opening up large gaps near the edges of the block. Data presented here show counterclockwise rotations of 16°, implying that much smaller interior rift blocks are rotating, perhaps broadly analogous to ball bearings between sliding rigid plates [e.g., Beck, 1976]. It is interesting to note that the interpretation of magnetotelluric data by Hermance and Pedersen [1980] and repeated leveling data by Reilinger and York [1979] both suggest the existence of a shallow magma body beneath the Espanola basin. Thus it seems possible that small intrarift blocks that are detached at depth could be rotating in a counterclockwise sense due to left slip along the rift.

Conclusions

1. Paleomagnetic studies of Miocene-Pliocene volcanic rocks in the Espanola basin (Paliza Canyon Formation, Tschicoma Formation, Lobato Basalt, and Cerros del Rio volcanics) yield stable paleomagnetic directions. All are close to, but distinct from, the expected direction for North America in the late Tertiary.

2. Basalts within the Santa Fe Group, located along the northeastern extension of the Pajarito fault zone, appear to have undergone a complex tilting and rotation history. Paleomagnetic data from these sites become widely scattered after a simple tilt correction.

3. Volcanic units studied here (Paliza Canyon Formation, Tschicoma Formation, and Lobato Basalt), lying west of the Pajarito fault zone, and published data on the Valles Rhyolite all have declinations east of the expected mean. A statistically significant positive rotation of 12° for this block indicates clockwise rotation for the western part of the Espanola basin.

4. Published data on the Tesuque Formation and Chamita Formation of the Santa Fe Group plus data presented here for the Cerros del Rio volcanics show declinations west of the expected mean. Counterclockwise motion of 16° is seen for this block east of the Pajarito fault zone.

5. Differential rotations between units of similar age on opposite sides of the Pajarito fault zone range from 25° to 31° and are statistically significant. Older units have greater rotations. If rotation began at approximately 5 Ma with the initiation of the Pajarito fault zone, then the minimum rate of differential rotation is roughly 4°/m.y.

6. Clockwise and counterclockwise rotations of different blocks within the Espanola basin can be related to extension and left slip along the Rio Grande rift. Clockwise motion of the western "Jemez" block may be in response to the opening of the rift about a pole in northern Colorado and the clockwise rotation of the Colorado Plateau or the rotation of small blocks in a WNW-ESE extensional stress field. Counterclockwise rotation of the eastern "Espanola" block of the Espanola basin is consistent with left slip along the Rio Grande rift, proposed from offset in Mesozoic units, drag along border faults, and compressional features north of the study area. The ease of rotation of these crustal blocks may be enhanced by shallow magma bodies beneath both the Jemez Mountains and the Espanola basin.

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L. L. Brown, Department of Geology and Geography, University of Massachusetts, Amherst, MA 01003.

M. P. Golombek, Mail Stop 183-501, Jet Propulsion Laboratory, California Institute of Technology, 4800 Oak Grove Drive, Pasadena, CA 91109.

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